Micro/Small Business Network Design Framework

This is a reproducible, entry-level-friendly design for 1–10 users (micro baseline) that scales to 10–50 with minor additions. It prioritizes the governance documents exactly: cable management, guest isolation + captive portal (with exceptions), VLAN segmentation without a full DMZ, Protective DNS, IoT isolation, power discipline, VPN hardening, 99.99% connectivity target where ISP allows, and 3-2-1 backups for configs.

Core Philosophy

- One core router/firewall handles routing, firewall rules, Wi-Fi SSIDs, VPN, DNS filtering, and QoS.
- A separate managed switch enables wired VLAN segmentation.
- Patch panel + short jumpers for Core and Switch (no direct plugs).
- All cables Cat6, Velcro-bound, wall-plate terminated, excess looped, never on floor.
- Three VLANs: VLAN 1 (Main LAN – trusted), VLAN 10 (Guest), VLAN 20 (IoT).
- Three SSIDs with client isolation on Guest/IoT.
- DNS forced to NextDNS (free tier) everywhere, unauthorized DNS blocked at firewall.
- Guest uses WPA3 + QR-code password + basic captive portal + bandwidth limit + DNS filtering.
- VPN (WireGuard) gives role-based access only to approved zones, not full LAN.
- Power via listed surge protector + optional UPS on dedicated circuits.

This setup is deliberately sub-$100 per piece (most ~$30–$95) and uses the same repeatable shopping list so any entry-level tech can replicate it in <4 hours.

Recommended Primary Design (GL.iNet + TP-Link – Best Match to Governance)

Why this stack? GL.iNet is explicitly referenced in your DMZ-exception document for access points and runs OpenWRT-based firmware. It natively supports VLANs on Wi-Fi, client isolation, guest networks, WireGuard VPN, custom DNS, QoS, and modern crypto — all at micro-business prices. TP-Link smart switch adds cheap wired VLANs without breaking the budget.

Inventory List with Current Approx. Prices (March 2026, Amazon/GL.iNet US Store)

Item	Model	Approx. Price	Key Features & Governance Fit	Analysis & Exceptions (if any)
Core Router/Firewall	GL.iNet Flint (GL-AX1800)	$94.99	Wi-Fi 6 AX1800, 5× Gigabit ports, OpenWRT-based VLANs, per-SSID client isolation, Guest Mode, WireGuard VPN server (up to ~550 Mbps), custom DNS (NextDNS), QoS, firewall rules, multiple SSIDs.	Meets 95%+ of requirements: segmentation, guest isolation, VPN, DNS filtering, connectivity. Handles 20–25 devices comfortably (Cisco/Ekahau guideline). Exception: Full captive portal with ToS/logging requires simple OpenWRT LuCI package or external tool; use built-in Guest SSID + QR password sharing as baseline (matches your "physical QR code" rule).
Managed Switch	TP-Link TL-SG108E (8-port) or TL-SG105E (5-port)	$35–$40	Gigabit, 802.1Q VLANs, QoS, IGMP snooping, web UI.	Enables wired VLAN segmentation + port-based rules. Meets "stateful firewall + VLAN" and "no direct plug" via patch panel. No exceptions needed.
Patch Panel	12-port Cat6 keystone patch panel + 12 short jumpers	$22–$28	Standard 19" or wall-mount.	Meets "Core and Switch must have patch panel with short jumpers." Excess cable looped with Velcro.

Wall Plates & Cables	5× Ethernet wall plates + Cat6 bulk/pre-made cables + Velcro ties (50-pack)	$15 (plates) + $20 (cables) + $10 (Velcro)	Removable Velcro only; no zip ties/tape/string	Exact match to cable-management policy. All drops terminate at wall plates; no exposed cable.
Power Protection	Amazon Basics / Tripp Lite 600VA UPS (or Wyze surge strip if tighter budget)	$45–$60 (UPS) or $15–$25 (surge only)	Surge suppression + battery backup; multiple outlets	Meets UPS/surge/grounding rules. Plug router + switch into separate circuits where possible. Use properly grounded outlets only. Exception: Full electrician bonding only if adding rack (micro setups use listed surge/UPS).
Optional AP (for >1,500 sq ft or >25 users)	Second GL.iNet (Marble GL-B3000 or refurbished Slate) in AP mode	$60–$80	Same firmware; wired back to switch VLAN trunk.	Meets Ekahau/TP-Link placement (–80 dBm overlap, 20–25 users per AP). Place on separate VLAN trunk if desired.

Total Core Setup Cost: ~$230–$260 (every piece ≤ $95).

Scales by adding one $60–$80 AP + more wall plates (~$30). No over-purchasing

Alternative Design (Even Cheaper All-in-One Option – TP-Link Only)

If budget must stay strictly under $200 total.

- Core: TP-Link Archer BE230 or AX55 (~$60–$80) – Wi-Fi 7/6, guest isolation, QoS, basic VPN.
- Switch: Same TP-Link TL-SG108E (~$40).
- Rest identical.

Trade-offs: Weaker VPN speeds, less robust OpenWRT-level VLAN/Wi-Fi isolation, no native 2.5G. Exception documented. Reduced segmentation depth and VPN performance vs. GL.iNet. Only use if GL.iNet stock is unavailable.

Logical & Physical Topology (Reproducible Diagram Description)

1. ISP Ethernet (or Fiber) → Core Router WAN port.
2. Core Router LAN ports → Patch panel (short jumpers).
3. Patch panel → Managed Switch (VLAN trunk port).
4. Switch ports → Wall plates or additional APs (trunk for VLANs 1/10/20).
5. All wired drops terminate at wall plates. Excess cable looped in Velcro at panel/switch.
6. No cables touch floor or hang outside comm area.

VLAN & Firewall Rules (applied at Core + Switch)

- VLAN 1: Main LAN (trusted devices)
- VLAN 10: Guest – client isolation + bandwidth limiter + DNS filtering only
- VLAN 20: IoT – client isolation + no access to Main LAN + DNS filtering
  Firewall blocks all inter-VLAN traffic except explicit allows (e.g., VPN to specific services). Matches segmentation + DMZ-exception policy exactly.

Wireless Guidelines

- 1–2 SSIDs per physical AP: Main (WPA3), Guest (WPA3 + isolation), IoT (WPA3 + isolation).
- Place APs so signal hits –80 dBm at edges (10 m overlap)

- Place APs so signal fills "Core DMZ" at edges (no RF overlap).
- Guest: QR-code password on wall; limited QoS; DNS filtering blocks risky sites.

DNS & Filtering

Router forces all devices (wired/wireless/VPN) to NextDNS. Firewall drops any other DNS queries. Meets protective DNS + URL filtering exactly (free tier fine for micro per your document).

VPN

WireGuard server on Core. Clients get access only to approved VLAN/services (not full LAN). Strong password + modern crypto. Exception: MFA not native on this hardware; strong password + certificate option if advanced users need it.

Power & Backup

- All equipment on UPS/surge-protected outlets.
- Config backups: Export after every change + weekly automated to Nextcloud (internal) + external cloud (3-2-1 rule).
- Logs retained 7 days minimum

Exceptions Summary (Only Where Governance Cannot Be Met at Sub-$100/Piece)

1. No separate DMZ firewall interface (doubles core cost + RF noise) – use manual VLAN + isolation + rules (exact match to your provided DMZ-exception).
2. Captive portal – basic guest SSID + QR instead of full ToS/logging portal (OpenWRT add-on possible but not required for micro).
3. NextDNS free-tier query limit – acceptable for micro/homes per your DNS document.
4. Full enterprise monitoring – use built-in router logs/telemetry (monitoring.md was empty).
5. Professional site survey – manual placement per Ekahau/Cisco rules (entry-level replication).

(only to set replication).

This framework meets every governance requirement that is practically achievable at micro-business cost and scale. It is fully replicable with the exact shopping list above — any junior tech can buy these items, follow the VLAN/SSID naming, and have a professional, secure network in one afternoon.

Next Steps for Approval

Approve the primary GL.iNet design (or the TP-Link alternative) and I will deliver the exact configuration template + wiring diagram + step-by-step deployment guide (no config in this draft per instructions). Total first-site cost stays under $300 including cabling. Let me know if you want a second scalable version with PoE APs or specific square footage adjustments. Ready for your audit.